

Sujeet Samal

B.Tech Aerospace Engineering (4th Semester)

CGPA (3rd Semester): 8.54/10.0

techsujetsamal@gmail.com | +91-6370327599 | LinkedIn: sujeet-samal-3629373a1 | GitHub: sujeetsamal

PROFESSIONAL SUMMARY

Aerospace Engineering undergraduate with practical experience in UAV systems, tilt-rotor VTOL configuration, flight controller integration, and embedded control architectures. Hands-on exposure to propulsion systems, power distribution, and ESP32-based IoT robotics. Strong foundation in aerodynamics, power systems, and control fundamentals with focus on autonomous flight technologies.

TECHNICAL SKILLS

Programming: Python, C, C++, JavaScript, HTML, CSS

Flight Systems: Flight Controller Integration, VTOL Configuration, ESC Calibration, Power Distribution

CAD: Fusion 360, CATIA, MATLAB

Embedded Systems: IOT

TECHNICAL PROJECTS

- **Tilt-Rotor VTOL Tricopter RC Plane:** Designed a 1.2 m wingspan, 1.5 kg tricopter VTOL system using 2×1000KV and 1×1400KV motors with 30A ESCs and 5000 mAh LiPo battery integrated with Pixhawk flight controller. Focused on tilt-rotor transition mechanism for hover-to-forward flight. Major engineering challenge involved mechanical tilt design and calibration stability.
- **S330 Quadcopter with Payload Capability:** Built a 1 kg quadcopter using 920KV motors, 30A ESCs, and 4200 mAh LiPo battery achieving 10 minutes flight time. Integrated radio link system and optimized thrust-to-weight ratio for stable payload operation.
- **Fixed-Wing RC Aircraft (JoyTrainer and Cessna Model):** Constructed 1.2 m wingspan trainer and Cessna style aircraft using Sunpack and Depron foam with NACA 4420 and standard Cessna airfoil profiles. Powered by 900KV motor and 2200 mAh LiPo battery achieving 6–8 minutes-controlled flight performance.
- **Helicopter Main Rotor Lab Prototype:** Developed an 80 cm diameter fixed-pitch rotor test rig using high-torque Johnson motor (60 RPM), L298N driver, and ESP32 Wi-Fi control system. Power supplied via 3×3.7V cells in series with regulated buck converter. Built as an experimental rotor propulsion model for lab study.
- **ESP32-Based Pole Climbing Robot:** Designed a three-actuator climbing mechanism with linear motion and dual gripping system controlled via hard-coded AI logic. Successfully demonstrated vertical pole climbing using actuator-sequenced control architecture.
- **LiDAR Object Detection Prototype:** Implemented VL53L0X Time-of-Flight sensor (2 m range) integrated with ESP32, and 180° servo sweep scanning to create a basic object detection and scanning system.
- **IoT Water Monitoring System:** Developed ESP-based Wi-Fi enabled & Solar powered water level monitoring system with LED indicators and buzzer alerts providing real-time level detection and remote-control capability.
- **ESP-Based RC Car (Class X Project):** Early embedded systems project demonstrating wireless motor control and ESP-based drive system.
- **Hybrid Electric Aircraft simulation:** A system-level hybrid-electric aircraft modeling project integrating propulsion simulation, flight dynamics, CAD modeling, and structural validation.

EDUCATION

B.Tech in Aerospace Engineering

Centurion University of Technology and Management, Bhubaneswar (Jatani Campus)

Current Semester: 4th

Class XII: Salipur Public School

Class X: St.John's School